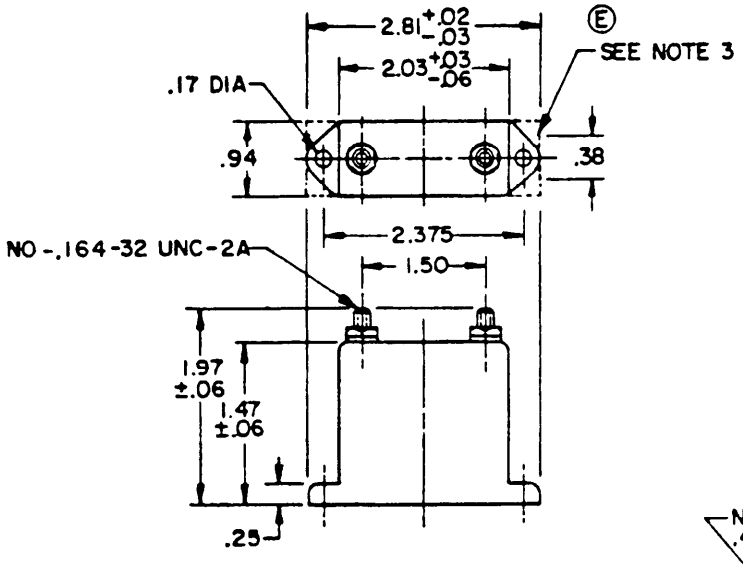
MIL-C-5/11E
5 September 1972
SUPERSEDING
MIL-C-5/11D
20 September 1966

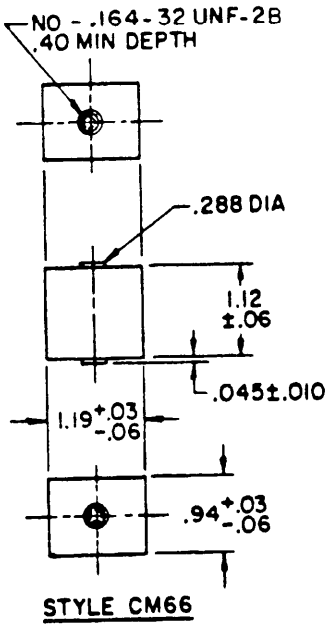
MILITARY SPECIFICATION SHEET
CAPACITORS, FIXED, MICA DIELECTRIC, STYLES CM65, CM66, AND CM68

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-5.



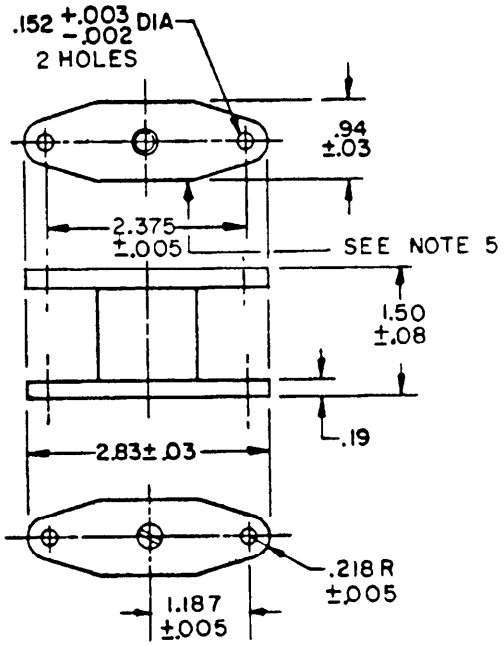
STYLE CM65



STYLE CM66

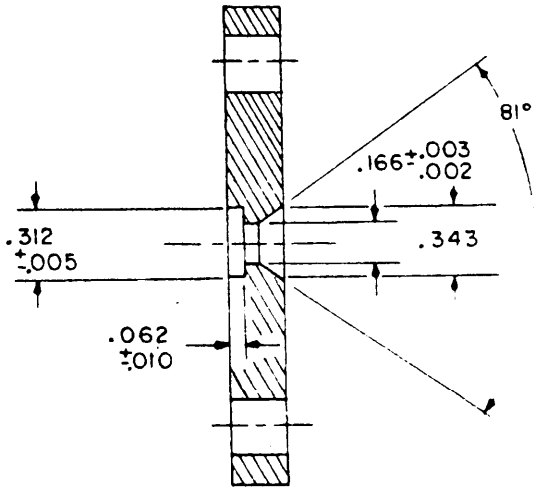
FSC 5910

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STYLE CM68

INCHES	MM
.002	.05
.003	.08
.005	.13
.010	.25
.02	.51
.03	.76
.045	1.14
.06	1.52
.062	1.57
.08	2.03
.152	3.86
.164	4.17
.166	4.22
.17	4.32
.19	4.83
.218	5.54
.25	6.35
.288	7.32
.312	7.92
.343	8.75
.40	10.16
.94	23.88
1.12	28.45
1.187	30.15
1.19	30.23
1.31	33.27
1.47	37.34
1.50	38.10
1.97	50.04
2.03	51.56
2.375	60.33
2.38	60.45
2.81	71.37
2.83	71.88



NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Dotted shape of mounting feet optional.
4. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and .02(.51 mm) for two place decimals.
5. Brackets (flanges) shall be removable.
6. Hardware to be plated against corrosion.
7. Approximate weight CM65 is .28 lbs. (127.58 g)

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REQUIREMENTS

Dimensions: See figure.

Material: Bracket (flange) - Aluminum per QQ-A-250/4 (T3-temper).

Construction: CM65 - Molded case.
CM66 and CM68 - Cast epoxy case.

Capacitance value: See table I.

Capacitance tolerance: ±2% (symbol G) and ±5% (symbol J).

Peak working voltage: See table I.

Operating temperature range:

Symbol	Temperature range
M	-55° C to +70° C
O	-55° C to +125° C

Vibration grade: Grade 1 (10-55 Hz).

Characteristic: See table I.

Symbol	Temperature coefficient P/M/° C	Capacitance drift
B	Not rated	Not rated
C	-200 to +200	±(0.5 percent +0.1 pF)
D	-100 to +100	±(0.3 percent +0.1 pF)
E	-20 to +100	±(0.1 percent +0.1 pF)
F	0 to +70	±(0.05 percent +0.1 pF)

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
AC sine-wave potential rms equal to peak working voltage at 100 cycles or less for not less than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202: condition B (50,000 ft).
Test potential: Capacitors rated above 1,000 volts - 130 percent of rated peak voltage. Capacitors rated at 1,000 volts or less - 150 percent of rated peak voltage.

Insulation resistance:
Method 302 of MIL-STD-202, condition A.
At 25° C - 7,500 megohms minimum for style CM65.
10,000 megohms minimum for styles CM66 and CM68.
At 70° C (CM65)
0 to 5,000 pF---2,000 megohms minimum.
greater than 5,000 pF---10 megohm microfarads.
At 125° C (CM66 and CM68)
0 to 5,000 pF---3,000 megohms, minimum.
greater than 5,000 pF---15 megohm microfarads.

Dissipation factor, maximum - See MIL-C-5.

Capacitance:
Method 305 of MIL-STD-202.

Current rating:
Passage of rated current shall not cause the temperature of the capacitor body's external surface to rise above the ambient temperature by more than 15° C for style CM65 and not more than 40° C for styles CM66 and CM68.

Vibration:
Method 201 of MIL-STD-202.

Temperature cycling:
Method 102 of MIL-STD-202, condition D, except the high temperature shall be as follows:
Symbol M - +70° C
Symbol O - +125° C

Immersion cycling:
Method 104 of MIL-STD-202, condition B.
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - 3,000 megohms minimum.
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±4 percent of nominal value or 1 pF, whichever is greater.

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Shock (specified pulse):
Method 213 of MIL-STD-202, condition I (100G(pk)).

Moisture resistance:
Method 106 of MIL-STD-202:
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±5 percent of nominal value or 1 pF, whichever is greater for characteristic B; ±2 percent of nominal value or 1 pF, whichever is greater for characteristics C, D, E and F.

Temperature coefficient and capacitance drift:
Within the limits specified for the applicable characteristic.

Low temperature storage and life:
48 hours at -55 $^{+0°}_{-2}$ C immediately followed by qualification or performance test, as applicable.
Qualification test - 1,000 hours at the applicable high test temperature with rated current at 1 MHz.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - Initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±8 percent of nominal value or 1 pF, whichever is greater, for characteristic B; ±3 percent of nominal value or 1 pF, whichever is greater, for characteristics C, D, E and F.

Performance check - 250 hours at conditions specified for qualification test.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - Initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement
Capacitance change - Shall not exceed ±5 percent of nominal value or 1 pF, whichever is greater, for characteristic B; ±2 percent of nominal value or 1 pF, whichever is greater, for characteristics C, D, E and F.

Continuation test - Units that have been subjected to the 250 hour test shall be continued for an additional 750 hours and meet the requirements for the qualification test. (This test not applicable to styles CM66 and CM68.)

Qualification inspection:
Qualification inspection (Group D) is required.

Supersession data:
Styles CM67 (all characteristics), CM66 and CM68 (characteristics B, C and F) have not been qualified; therefore, no interchangeability substitution data are offered.

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TABLE I. Styles CM65, CM66, and CM68.

Type designation 1/	Capacitance value (pF)	Peak working voltage (volts)	Rated current at frequencies (rms amps)				Characteristic
			3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	
CM65-L470-M1	47	3,000	1.2	.51	.15	.051	B, D, F
CM65-L510-M1	51	3,000	1.3	.51	.18	.056	
CM65-L560-M1	56	3,000	1.3	.56	.20	.056	
CM65-L620-M1	62	3,000	1.5	.62	.22	.068	
CM65-L680-M1	68	3,000	1.5	.62	.24	.075	
CM65-L750-M1	75	3,000	1.5	.62	.27	.082	
CM65-L820-M1	82	3,000	1.6	.68	.27	.082	
CM65-L910-M1	91	3,000	1.6	.68	.33	.091	
CM65-L101-M1	100	3,000	1.8	.75	.33	.10	
CM65-L111-M1	110	3,000	1.8	.82	.36	.12	
CM65-L121-M1	120	3,000	2.0	.91	.39	.15	
CM65-L131-M1	130	3,000	2.0	.91	.43	.16	
CM65-L151-M1	150	3,000	2.2	1.0	.47	.18	
CM65-L161-M1	160	3,000	2.2	1.1	.51	.20	
CM65-L181-M1	180	3,000	2.4	1.1	.56	.22	
CM65-L201-M1	200	3,000	2.4	1.2	.62	.24	
CM65-L221-M1	220	3,000	2.7	1.3	.62	.27	
CM65-L241-M1	240	3,000	2.7	1.3	.68	.30	
CM65-L271-M1	270	3,000	3.0	1.5	.75	.30	
CM65-L301-M1	300	3,000	3.0	1.5	.75	.36	
CM65-L331-M1	330	3,000	3.0	1.6	.82	.36	
CM65-L361-M1	360	3,000	3.3	1.6	.91	.39	
CM65-L391-M1	390	3,000	3.3	1.8	.91	.43	
CM65-L431-M1	430	3,000	3.6	1.8	1.0	.47	
CM65-L471-M1	470	3,000	3.6	2.0	1.1	.47	
CM65-L511-M1	510	3,000	3.6	2.0	1.1	.51	
CM65-L561-M1	560	3,000	3.9	2.2	1.2	.51	
CM65-L621-M1	620	3,000	3.9	2.4	1.2	.56	
CM65-L681-M1	680	3,000	4.3	2.4	1.3	.62	
CM65-L751-M1	750	3,000	4.3	2.7	1.3	.62	
CM65-L821-M1	820	3,000	4.3	2.7	1.5	.68	
CM65-L911-M1	910	3,000	4.7	3.0	1.5	.68	
CM65-L102-M1	1,000	3,000	4.7	3.0	1.6	.75	
CM65-L112-M1	1,100	3,000	5.1	3.3	1.6	.82	
CM65-L122-M1	1,200	3,000	5.1	3.3	1.8	.82	
CM65-L132-M1	1,300	3,000	5.6	3.6	1.8	.91	
CM65-L152-M1	1,500	3,000	5.6	3.9	2.0	.91	
CM65-L162-M1	1,600	3,000	5.6	3.9	2.0	1.0	
CM65-L182-M1	1,800	3,000	6.2	4.3	2.2	1.1	
CM65-L202-M1	2,000	3,000	6.2	4.3	2.4	1.1	
CM65-L222-M1	2,200	3,000	6.8	4.7	2.4	1.2	
CM65-L242-M1	2,400	3,000	6.8	4.7	2.7	1.2	
CM65-J272-M1	2,700	2,000	6.8	5.1	2.7	1.3	
CM65-J302-M1	3,000	2,000	7.5	5.1	3.0	1.3	
CM65-J332-M1	3,300	2,000	7.5	5.6	3.0	1.5	
CM65-J362-M1	3,600	2,000	7.5	5.6	3.3	1.5	
CM65-J392-M1	3,900	2,000	8.2	6.2	3.3	1.6	
CM65-J432-M1	4,300	2,000	8.2	6.2	3.6	1.6	
CM65-J472-M1	4,700	2,000	8.2	6.8	3.6	1.8	
CM65-J512-M1	5,100	2,000	8.2	6.8	3.6	1.8	
CM65-J562-M1	5,600	2,000	9.1	7.5	3.9	2.0	
CM65-J622-M1	6,200	2,000	9.1	7.5	4.3	2.0	
CM65-J682-M1	6,800	2,000	10	7.5	4.3	2.0	
CM65-J752-M1	7,500	2,000	10	8.2	4.7	2.2	B, D, F

See footnotes at end of table.

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TABLE I. Styles CM65, CM66, and CM68. - Continued

Type designation 1/2/	Capacitance value (pF)	Peak working voltage (volts)	Rated current at frequencies (rms amps)				Characteristic
			3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	
CM65-H822-M1	8,200	1,500	10	8.2	4.7	2.2	B, D, F
CM65-H912-M1	9,100	1,500	10	8.2	5.1	2.4	
CM65-F103-M1	10,000	1,000	10	9.1	5.1	2.4	
CM65-F113-M1	11,000	1,000	11	10	5.6	2.7	
CM65-F123-M1	12,000	1,000	11	10	6.2	3.0	
CM65-F133-M1	13,000	1,000	11	11	6.8	3.0	
CM65-F153-M1	15,000	1,000	11	11	6.8	3.3	
CM65-F163-M1	16,000	1,000	11	11	6.8	3.3	
CM65-F183-M1	18,000	1,000	11	11	7.5	3.3	
CM65-F203-M1	20,000	1,000	11	11	7.5	3.6	
CM65-F223-M1	22,000	1,000	11	11	7.5	3.6	B, D, F
CM65-F243-M1	24,000	1,000	11	11	7.5	3.6	C, E
CM65-D273-M1	27,000	500	11	11	7.5	3.9	
CM65-D303-M1	30,000	500	11	11	8.2	3.9	
CM65-D333-M1	33,000	500	11	11	8.2	4.3	
CM65-D363-M1	36,000	500	11	11	8.2	4.3	
CM65-D393-M1	39,000	500	11	11	8.2	4.3	
CM65-D433-M1	43,000	500	11	11	9.1	4.7	
CM65-B473-M1	47,000	250	11	11	9.1	4.7	
CM65-B513-M1	51,000	250	11	11	9.1	4.7	
CM65-B563-M1	56,000	250	11	11	9.1	4.7	
CM65-B623-M1	62,000	250	11	11	9.1	5.1	
CM65-B683-M1	68,000	250	11	11	9.1	5.1	
CM65-B753-M1	75,000	250	11	11	9.1	5.1	
CM65-B823-M1	82,000	250	11	11	9.1	5.1	
CM65-B913-M1	91,000	250	11	11	9.1	5.6	
CM65-B104-M1	100,000	250	11	11	9.1	5.6	C, E
CM6--L470-O1	47	3,000	1.2	.51	.15	.051	D, E
CM6--L510-O1	51	3,000	1.3	.51	.18	.056	
CM6--L560-O1	56	3,000	1.3	.56	.20	.056	
CM6--L620-O1	62	3,000	1.5	.62	.22	.068	
CM6--L680-O1	68	3,000	1.5	.62	.24	.075	
CM6--L750-O1	75	3,000	1.5	.62	.27	.082	
CM6--L820-O1	82	3,000	1.6	.68	.27	.082	
CM6--L910-O1	91	3,000	1.6	.68	.33	.091	
CM6--L101-O1	100	3,000	1.8	.75	.33	.10	
CM6--L111-O1	110	3,000	1.8	.82	.36	.12	
CM6--L121-O1	120	3,000	2.0	.91	.39	.15	
CM6--L131-O1	130	3,000	2.0	.91	.43	.16	
CM6--L151-O1	150	3,000	2.2	1.0	.47	.18	
CM6--L161-O1	160	3,000	2.2	1.1	.51	.20	
CM6--L181-O1	180	3,000	2.4	1.1	.56	.22	
CM6--L201-O1	200	3,000	2.4	1.2	.62	.24	
CM6--L221-O1	220	3,000	2.7	1.3	.62	.27	
CM6--L241-O1	240	3,000	2.7	1.3	.68	.30	
CM6--L251-O1	250	3,000	2.7	1.3	.68	.30	
CM6--L271-O1	270	3,000	3.0	1.5	.75	.30	
CM6--L301-O1	300	3,000	3.0	1.5	.75	.36	
CM6--L331-O1	330	3,000	3.0	1.6	.82	.36	
CM6--L361-O1	360	3,000	3.3	1.6	.91	.39	
CM6--L391-O1	390	3,000	3.3	1.8	.91	.43	
CM6--L431-O1	430	3,000	3.6	1.8	1.0	.47	
CM6--L471-O1	470	3,000	3.6	2.0	1.1	.47	
CM6--L501-O1	500	3,000	3.6	2.0	1.1	.51	
CM6--L511-O1	510	3,000	3.6	2.0	1.1	.51	
CM6--L561-O1	560	3,000	3.9	2.2	1.2	.51	
CM6--L621-O1	620	3,000	3.9	2.4	1.2	.56	D, E

See footnotes

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TABLE I. Styles CM65, CM66, and CM68. - Continued

Type designation 1/2/	Capacitance value (pF)	Peak working voltage (volts)	Rated current at frequencies (rms amps)				Characteristic
			3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	
CM6--L681-O1	680	3,000	4.3	2.4	1.3	.62	D, E
CM6--L751-O1	750	3,000	4.3	2.7	1.3	.62	
CM6--L821-O1	820	3,000	4.3	2.7	1.5	.68	
CM6--L910-O1	910	3,000	4.7	3.0	1.5	.68	
CM6--L102-O1	1,000	3,000	4.7	3.0	1.6	.75	
CM6--L112-O1	1,100	3,000	5.1	3.3	1.6	.82	
CM6--L122-O1	1,200	3,000	5.1	3.3	1.8	.82	
CM6--L132-O1	1,300	3,000	5.6	3.6	1.8	.91	
CM6--L152-O1	1,500	3,000	5.6	3.9	2.0	.91	
CM6--L162-O1	1,600	3,000	5.6	3.9	2.0	1.0	
CM6--L182-O1	1,800	3,000	6.2	4.3	2.2	1.1	
CM6--L202-O1	2,000	3,000	6.2	4.3	2.4	1.1	
CM6--L222-O1	2,200	3,000	6.8	4.7	2.4	1.2	
CM6--L242-O1	2,400	3,000	6.8	4.7	2.7	1.2	
CM6--J272-O1	2,700	2,000	6.8	5.1	2.7	1.3	
CM6--J302-O1	3,000	2,000	7.5	5.1	3.0	1.3	
CM6--J332-O1	3,300	2,000	7.5	5.6	3.0	1.5	
CM6--J362-O1	3,600	2,000	7.5	5.6	3.3	1.5	
CM6--J392-O1	3,900	2,000	8.2	6.2	3.3	1.6	
CM6--J402-O1	4,000	2,000	8.2	6.2	3.3	1.6	
CM6--J432-O1	4,300	2,000	8.2	6.2	3.6	1.6	
CM6--J472-O1	4,700	2,000	8.2	6.8	3.6	1.8	
CM6--J502-O1	5,000	2,000	8.2	6.8	3.6	1.8	
CM6--J512-O1	5,100	2,000	8.2	6.8	3.6	1.8	
CM6--J562-O1	5,600	2,000	9.1	7.5	3.9	2.0	
CM6--J602-O1	6,000	2,000	9.1	7.5	4.3	2.0	
CM6--J622-O1	6,200	2,000	9.1	7.5	4.3	2.0	
CM6--J682-O1	6,800	2,000	10.0	7.5	4.3	2.0	
CM6--J752-O1	7,500	2,000	10.0	8.2	4.7	2.2	
CM6--H802-O1	8,000	1,500	10.0	8.2	4.7	2.2	
CM6--H822-O1	8,200	1,500	10.0	8.2	4.7	2.2	
CM6--H912-O1	9,100	1,500	10.0	8.2	5.1	2.4	
CM6--F103-O1	10,000	1,000	10.0	9.1	5.1	2.4	
CM6--F113-O1	11,000	1,000	11.0	10.0	5.6	2.7	
CM6--F123-O1	12,000	1,000	11.0	10.0	6.2	3.0	
CM6--F133-O1	13,000	1,000	11.0	11.0	6.8	3.0	
CM6--F153-O1	15,000	1,000	11.0	11.0	6.8	3.3	
CM6--F163-O1	16,000	1,000	11.0	11.0	6.8	3.3	
CM6--F183-O1	18,000	1,000	11.0	11.0	7.5	3.3	
CM6--F203-O1	20,000	1,000	11.0	11.0	7.5	3.6	
CM6--F223-O1	22,000	1,000	11.0	11.0	7.5	3.6	
CM6--F243-O1	24,000	1,000	11.0	11.0	7.5	3.6	
CM6--D273-O1	27,000	500	11.0	11.0	7.5	3.9	
CM6--D303-O1	30,000	500	11.0	11.0	8.2	3.9	
CM6--D333-O1	33,000	500	11.0	11.0	8.2	4.3	
CM6--D363-O1	36,000	500	11.0	11.0	8.2	4.3	
CM6--D393-O1	39,000	500	11.0	11.0	8.2	4.3	
CM6--D433-O1	43,000	500	11.0	11.0	9.1	4.7	
CM6--B473-O1	47,000	250	11.0	11.0	9.1	4.7	
CM6--B503-O1	50,000	250	11.0	11.0	9.1	4.7	
CM6--B513-O1	51,000	250	11.0	11.0	9.1	4.7	
CM6--B563-O1	56,000	250	11.0	11.0	9.1	4.7	
CM6--B623-O1	62,000	250	11.0	11.0	9.1	5.1	
CM6--B683-O1	68,000	250	11.0	11.0	9.1	5.1	
CM6--B753-O1	75,000	250	11.0	11.0	9.1	5.1	
CM6--B823-O1	82,000	250	11.0	11.0	9.1	5.1	
CM6--B913-O1	91,000	250	11.0	11.0	9.1	5.6	
CM6--B104-O1	100,000	250	11.0	11.0	9.1	5.6	D, E

1 For style CM6-- the complete type designation will include additional symbols to indicate characteristic and capacitance tolerance.

2 For styles CM65 and CM68, the complete type designation will include additional symbols to indicate style, characteristic, and capacitance tolerance.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy - AS, OS, EC, SH
Air Force - 11, 17,
DSA - ES

User activities:
Army - ME
Navy - MC, CG
Air Force - 19

Preparing activity:
Air Force - 80

Agent:
DSA - ES

(Project 5910-1044-1)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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